

ON THE PITFALLS OF VERBALIZED CONFIDENCE PRIORS FOR CALIBRATING REASONING LLMs

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Paper under double-blind review

ABSTRACT

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1 INTRODUCTION

Reasoning large language models (LLMs) increasingly solve complex problems by generating a reasoning trajectory followed by the answer (Yang et al., 2025). Despite these advances, the answer accuracy alone does not ensure reliable uncertainty estimates: a model may produce plausible reasoning trajectory yet remain overconfident in an incorrect answer (Geng et al., 2024; Liu et al., 2025). In principle, confidence calibration requires the reported confidence to reflect the empirical probability that the answer is correct, which in turn enables applications such as hallucination detection (Wu et al., 2026) and adaptive test-time scaling (Yang et al., 2026).

Verbalized confidence obtains such an answer-level score by asking the model to emit a numerical probability with its final answer. Unlike confidence estimators based on token probabilities, hidden states, or repeated sampling (Luo et al., 2026; Srey et al., 2026; Wang & Stengel-Eskin, 2026), it can be produced in a single black-box generation (Xiong et al., 2024).

2 PRELIMINARIES

2.1 VERBALIZED CONFIDENCE AND CONFIDENCE-AWARE RL

In this work, we study the calibration of verbalized numerical confidence in reasoning language models (Tian et al., 2023; Xiong et al., 2024). Given a question x , the LLM π_θ generates a response $o = (r, a, c)$, where r is a reasoning trace, a is an answer, and $c \in [0, 1]$ is a post-answer confidence value that represents the model’s estimated probability that a is correct. A task verifier assigns the response a binary correctness label $z = V(x, a) \in \{0, 1\}$.

Confidence-aware rewards. Reinforcement learning with verifiable rewards (RLVR) trains the policy with the correctness label z as the reward. To further incorporate confidence into the optimization objective, confidence-aware RL adds a calibration reward that depends on both c and z . For example, RLCR (Damani et al., 2026) sets the calibration reward to the negative Brier loss. The resulting total reward favors correct answers and, for a given correctness, prefers confidence closer to z :

$$R(o) = z - (c - z)^2. \quad (1)$$

Policy optimization. Confidence-aware RL typically optimizes this reward with Group Relative Policy Optimization (GRPO) (Shao et al., 2024). For each question, GRPO samples G responses $\{o_i\}_{i=1}^G$ from the old policy $\pi_{\theta_{\text{old}}}$ and normalizes their rewards $R_i = R(o_i)$ within the group to obtain the advantage

$$\hat{A}_i = \frac{R_i - \bar{R}}{\sigma_R + \epsilon}, \quad \bar{R} = \frac{1}{G} \sum_{j=1}^G R_j, \quad \sigma_R = \text{std}(\{R_j\}_{j=1}^G), \quad (2)$$

where $\epsilon > 0$ ensures numerical stability. GRPO then updates the policy by maximizing

$$J_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{L_i} \sum_{t=1}^{L_i} \min \left(\rho_{i,t}(\theta) \hat{A}_i, \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_i \right) \right], \quad (3)$$

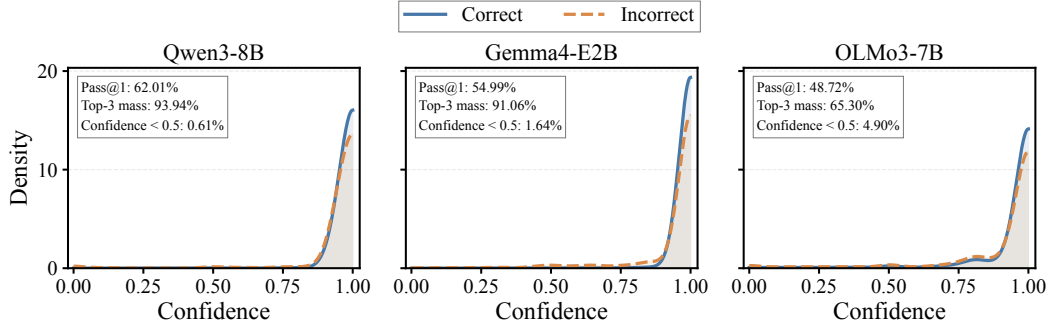


Figure 1: Confidence distributions from three base models on DeepScaleR-Train. Blue denotes correct responses and orange denotes incorrect responses. Confidence concentrates on a few high values for both correct and incorrect responses.

where $o_{i,t}$ is the t -th token of o_i , L_i is the length of o_i , $\rho_{i,t}(\theta) = \pi_\theta(o_{i,t} \mid x, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid x, o_{i,<t})$ is the importance ratio, and ϵ_{clip} is the clipping parameter.

2.2 CALIBRATION METRICS

A useful verbalized confidence score should be both calibrated and discriminative (Eusebi et al., 2026). Let C and Z denote the confidence and correctness random variables. Confidence is perfectly calibrated if $\Pr(Z = 1 \mid C = p) = p$ for every $p \in [0, 1]$, and it is discriminative if correct responses receive higher confidence than incorrect ones.

Expected Calibration Error (ECE) (Guo et al., 2017) measures the deviation from perfect calibration, $\mathbb{E}_C[|\Pr(Z = 1 \mid C) - C|]$. After partitioning N responses into B confidence bins $\{\mathcal{I}_b\}_{b=1}^B$, we estimate it as

$$\text{ECE} = \sum_{b=1}^B \frac{|\mathcal{I}_b|}{N} |\text{acc}(\mathcal{I}_b) - \text{conf}(\mathcal{I}_b)|, \quad (4)$$

where $\text{acc}(\mathcal{I}_b)$ and $\text{conf}(\mathcal{I}_b)$ are the mean correctness and confidence in bin b . While ECE compares confidence with accuracy at the bin level, the **Brier score** (Brier, 1950), $\frac{1}{N} \sum_{j=1}^N (c_j - z_j)^2$, measures the squared error between confidence and correctness for each response and combines calibration and resolution (Murphy, 1973). However, calibration alone does not imply discrimination. For example, assigning every response a constant confidence equal to the overall accuracy yields zero ECE while giving correct and incorrect responses identical scores (Eusebi et al., 2026). We therefore also report **AUROC**, a threshold-independent measure of how well confidence discriminates between correct and incorrect responses.

3 MOTIVATION

Confidence-aware RL typically adds a calibration reward that scores each response’s confidence against its correctness (Xu et al., 2024; Damani et al., 2026; Ma et al., 2026; Yang et al., 2026). Whether this reward alone produces well-calibrated confidence depends on the distribution of sampled confidence values. We therefore examine this distribution before RL, during training, and at test time.

Confidence priors concentrate on a few high values. We refer to the confidence distribution before RL as the model’s *confidence prior*. To characterize this prior, we sample 10 responses per question from Qwen3-8B (Yang et al., 2025), Gemma4-E2B-Instruct (Abd et al., 2026), and OLMo3-7B-Instruct (Ettinger et al., 2025) on DeepScaleR-Train. Figure 1 presents their confidence distributions for correct and incorrect responses. All three models place most of their confidence mass on a few high values, with fewer than 5% of outputs below 0.5. Confidence remains high even for incorrect responses, so it does not track answer correctness. RL therefore starts from a concentrated confidence prior that rarely produces low-confidence samples.

RL improves calibration but confidence concentration persists. To examine whether confidence-aware RL overcomes this concentration, we train Qwen3-8B with RLCR (Damani et al., 2026). As shown in Figure 2(a), RLCR reduces Brier loss over training, showing that the calibration reward can be optimized. However, Figure 2(b) shows that the model continues to sample a small set of confidence values per question, and confidence token entropy stays low throughout training (see Appendix B for training curves across RL objectives).

To better understand why this concentration persists during training, we formulate confidence generation as action selection under a softmax policy (Garg et al., 2022). Since the LLM decodes each confidence value autoregressively, we model the induced distribution over the finite set of confidence values as a softmax over per-value logits.

Proposition 1 (Policy gradients under confidence concentration). *Given a question x and a finite set of confidence actions $\{c_1, \dots, c_K\} \subseteq [0, 1]$, let $\eta = (\eta_1, \dots, \eta_K)$ denote the policy’s logits for these actions and $\pi_i = \Pr(C = c_i \mid x) = \exp(\eta_i) / \sum_{j=1}^K \exp(\eta_j)$. Define the fixed expected reward $u_i = -\mathbb{E}[(c_i - Z)^2 \mid x]$. Let $J_x(\eta) = \sum_i \pi_i u_i$ and $\Delta_x = \max_i u_i - \min_i u_i$. Then*

$$\frac{\partial J_x}{\partial \eta_i} = \pi_i(u_i - J_x), \quad \left| \frac{\partial J_x}{\partial \eta_i} \right| \leq \pi_i \Delta_x. \quad (5)$$

If $\pi_k = 1 - \varepsilon$ for some $k \in \{1, \dots, K\}$, then

$$\|\nabla_{\eta} J_x\|_1 \leq 2\varepsilon \Delta_x. \quad (6)$$

Proposition 1 shows that a rarely sampled confidence value has a small expected logit gradient even when choosing it would improve the expected reward. As the confidence distribution concentrates on a single value, the entire logit gradient vanishes, including when that value is suboptimal. A concentrated confidence distribution therefore impedes exploration of better-calibrated outputs, consistent with the persistently low distinct-value count and confidence token entropy in Figure 2(b). We provide the proof in Appendix C.1.

Persistent confidence concentration hinders downstream calibration. We evaluate the final RLCR checkpoint on Math500 and MinervaMath, two datasets of contrasting difficulty. As shown in Figure 3, the post-RL model underestimates its accuracy on Math500 (90.5% Pass@1 vs. 84.5% mean confidence) but is overconfident on MinervaMath (36.7% Pass@1 vs. 74.8% mean confidence). Despite a 53.8-point gap in Pass@1 between the two tasks, mean confidence differs by only 9.7 points. The three most frequent values still account for 82.25% and 69.49% of outputs on Math500 and MinervaMath, respectively. The same small set of confidence values therefore persists across tasks of different difficulty, so confidence no longer tracks the model’s accuracy on each task.

To characterize the cost of this miscalibration, we bound the expected Brier loss $\mathbb{E}[(C - Z)^2 \mid x]$, which is minimized at $C = p_x$, where $p_x = \Pr(Z = 1 \mid x)$ (Gneiting & Raftery, 2007; Damani et al., 2026).

Proposition 2 (Brier risk under limited confidence support). *Assume that C and Z are conditionally independent given x , consistent with the limited discrimination between correct and incorrect responses observed in Figure 1. For any $\delta > 0$, let $m_{\delta}(x) = \Pr(|C - p_x| < \delta \mid x)$. Then*

$$\mathbb{E}[(C - Z)^2 \mid x] \geq p_x(1 - p_x) + \delta^2(1 - m_{\delta}(x)). \quad (7)$$

Proposition 2 shows that the lower bound on Brier loss increases as $m_{\delta}(x)$ decreases, so concentration hurts calibration whenever the retained values lie far from p_x . In Appendix C.2, we provide the detailed proof, extend the bound to a restricted confidence support, and show that limited support similarly raises the expected error of the confidence value closest to p_x among G samples.

4 METHOD

Section 3 showed that confidence concentration persists during RL, since rarely sampled confidence values have small logit gradients, and that the resulting narrow support hinders downstream calibration. Our key idea is therefore to set the confidence prior by supervision before RL, rather than leaving it to exploration. We propose CalibSFT, a supervised fine-tuning stage for confidence initialization.

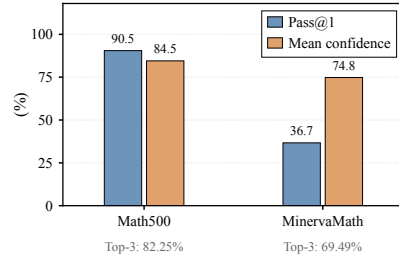
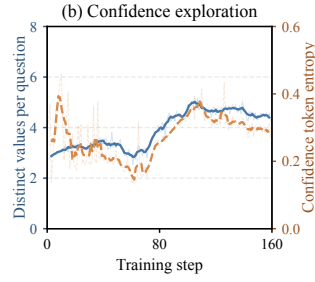
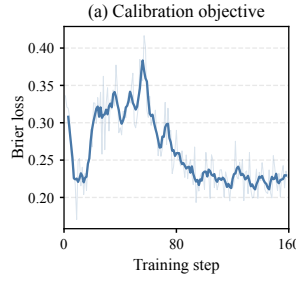


Figure 2: Training dynamics of RLCR on Qwen3-8B. (a) Brier loss. (b) Distinct confidence per question (blue) and confidence token entropy (orange). RLCR reduces Brier loss but confidence exploration remains narrow.

Figure 3: Performance comparison on Math500 and MinervaMath of RLCR on Qwen3-8B. Confidence differs little despite their large accuracy gap.

For each question, CalibSFT samples multiple responses from the base model and assigns each a confidence target that combines the question’s success rate with the response’s own correctness. It supervises confidence on all responses, but reasoning and answers only on correct ones.

4.1 PROPER-SCORING CONFIDENCE TARGETS

Under a proper scoring rule, the optimal confidence equals the probability that the answer is correct (Gneiting & Raftery, 2007). Since this probability is not observed directly, we estimate it from the base model’s own responses. For each training question x , we sample n responses with correctness labels $z_i \in \{0, 1\}$ and use their success rate as the estimate. However, correct and incorrect responses to the question then share the same confidence target. To support discrimination, we combine the success rate with the response-level label z_i and construct the confidence target of response i as

$$q_x = \frac{1}{n} \sum_{i=1}^n z_i, \quad c_i^* = \lambda q_x + (1 - \lambda) z_i, \quad (8)$$

where the mixing weight $\lambda \in [0, 1]$ balances the two signals. The mixed target is no longer a correctness label, so we examine whether it still yields the same optimum under a strictly proper scoring rule. Let $p_x = \Pr(Z = 1 \mid x)$ under the base model, and let Z_i and C_i^* denote the random counterparts of z_i and c_i^* when the n responses are sampled independently. We then have the following proposition.

Proposition 3 (Proper-scoring optimum of the mixed target). *Let $\ell(a, z)$ be a strictly proper scoring loss for $z \in \{0, 1\}$, extended to $t \in [0, 1]$ by $\ell(a, t) = t \ell(a, 1) + (1 - t) \ell(a, 0)$. Then $\mathbb{E}[C_i^* \mid x] = p_x$, and for every a at which ℓ is finite,*

$$\mathbb{E}[\ell(a, C_i^*) \mid x] = \mathbb{E}[\ell(a, Z_i) \mid x]. \quad (9)$$

Proposition 3 shows that the mixed target is equivalent to the correctness label in expected loss, so p_x remains the optimal prediction under any strictly proper ℓ . The number of responses n controls the variance of the success-rate estimate and the weight λ controls the target’s discrimination within a question; we set $n = 50$ and $\lambda = 0.5$ by default. We prove the proposition in detail and derive the variance of C_i^* in Appendix C.3.

In practice, we replace each response’s confidence with c_i^* , preserving its reasoning and answer. Since the target values are unevenly distributed across questions, training on the raw data would again concentrate supervision on a few target values. We therefore apply target-balanced sampling, which draws the same number of responses for each target value (Appendix D.1).

4.2 CORRECTNESS-CONDITIONAL SUPERVISION

The constructed data contain both correct and incorrect responses. The incorrect ones are kept because their lower targets teach the model to express low confidence in its own errors, whereas

imitating their reasoning would train the model on incorrect solutions. CalibSFT therefore supervises confidence on all responses and restricts reasoning and answer supervision to correct ones.

Formally, let $\tilde{o}$ denote a response whose confidence has been replaced by its constructed target, and let $\mathcal{T}_{\text{ans}}$ and $\mathcal{T}_{\text{conf}}$ denote the positions of its reasoning and answer tokens and its confidence tokens, respectively. The CalibSFT objective is then

$$\begin{aligned} \mathcal{L}_{\text{CalibSFT}} = & (1 - \beta) \mathbb{E}_{\text{correct}} \left[-\frac{1}{|\mathcal{T}_{\text{ans}}|} \sum_{t \in \mathcal{T}_{\text{ans}}} \log \pi_{\theta}(\tilde{o}_t \mid x, \tilde{o}_{<t}) \right] \\ & + \beta \mathbb{E}_{\text{all}} \left[-\frac{1}{|\mathcal{T}_{\text{conf}}|} \sum_{t \in \mathcal{T}_{\text{conf}}} \log \pi_{\theta}(\tilde{o}_t \mid x, \tilde{o}_{<t}) \right]. \end{aligned} \quad (10)$$

Here $\mathbb{E}_{\text{correct}}$ averages over the correct responses and $\mathbb{E}_{\text{all}}$ over all of them. Averaging within each segment prevents the longer reasoning sequences from dominating the loss by token count. The weight β balances confidence supervision against reasoning supervision, and we set $\beta = 0.5$ so that the two terms contribute equally.

Integration with confidence-aware RL. CalibSFT only changes the confidence prior and leaves the RL objective unchanged, so it can be applied before any confidence-aware RL method. We evaluate it across RL methods in Section 5.2 and analyze its effect on confidence concentration after RL in Section 5.3. We further examine its generality across confidence formats and base models in Section 5.4. Finally, we ablate the confidence target construction, the mixing weight λ , and the correctness-conditional supervision in Appendix E.2.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines. We compare methods spanning answer-only and confidence-aware training. **Base** denotes the checkpoint without task-specific training, while **RLVR** optimizes only answer correctness. We use confidence-aware RL baselines including **RLCR** (Damani et al., 2026), **ReDoubt** (Bani-Harouni et al., 2026), **CoCA** (Li et al., 2026a), **DCPO** (Ma et al., 2026), and **C3RL** (Yang et al., 2026). To measure the effect of **CalibSFT**, we evaluate each confidence-aware RL method from both the Base and CalibSFT checkpoints. All methods use Qwen3-8B (Yang et al., 2025) as the default model and follow the **DAPO** training recipe (Yu et al., 2025).

Datasets. We use **DeepScaleR-Preview** (Luo et al., 2025) as the mathematical training dataset. We evaluate on a broad range of in-distribution (ID) and out-of-domain (OOD) datasets spanning different levels of difficulty. For ID mathematical reasoning, we use **DeepScaleR-Eval**, **MATH-500** (Hendrycks et al., 2021), **MinervaMath** (Lewkowycz et al., 2022), **OlympiadBench** (He et al., 2024), **GSM8K** (Cobbe et al., 2021), and **AIME 2024–2026** (Dekoninck et al., 2026). For general reasoning, we use **HotpotQA** (Yang et al., 2018), **MuSiQue** (Trivedi et al., 2022), **TriviaQA** (Joshi et al., 2017), **NQOpen** (Lee et al., 2019), **PopQA** (Mallen et al., 2023), **WebQuestions** (Berant et al., 2013), **DROP** (Dua et al., 2019), and **LiveBenchReasoning** (White et al., 2024).

Evaluation metrics. We evaluate model performance using complementary metrics that capture answer correctness, confidence discrimination, and absolute calibration. We report **Pass@1** for answer correctness, **AUROC** for ranking correct responses above incorrect responses, and **Brier score** (Brier, 1950) and **ECE** (Guo et al., 2017) for probabilistic calibration.

Implementation details. During evaluation, we use temperature 0.6 and sample 32 responses per problem for the AIME benchmarks and 4 responses per problem for all other benchmarks. All experiments are conducted on a single node with eight NVIDIA A100 GPUs. We report our hyperparameter setting and training details in Appendix D.

Table 1: Performance comparison on in-distribution and out-of-distribution benchmarks. Each entry averages the datasets within the corresponding split. CalibSFT initialization achieves notably better calibration, discrimination, and Pass@1 for all RL baseline.

Method	In-distribution				Out-of-distribution			
	Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$	Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
Base	57.39	71.47	34.97	36.26	45.36	65.71	45.15	46.94
RLVR	61.60	73.16	33.48	34.08	43.88	64.19	50.29	51.52
CalibSFT	56.20	79.25	18.01	14.82	45.91	61.38	28.25	24.16
RLCR	62.71	82.78	15.60	16.84	43.96	69.38	31.14	31.17
+ CalibSFT	64.70	86.88	13.48	10.84	45.71	71.31	23.24	18.20
ReDoubt	63.08	81.70	15.94	18.85	43.71	69.88	25.07	23.98
+ CalibSFT	64.99	88.33	13.00	16.20	45.55	73.93	22.42	18.89
CoCA	61.11	80.59	17.25	21.75	43.68	66.79	26.67	24.12
+ CalibSFT	62.65	86.31	16.40	20.21	45.55	71.21	22.95	18.89
DCPO	61.98	78.16	16.38	14.68	43.99	68.13	32.85	32.41
+ CalibSFT	62.40	85.16	13.29	12.20	44.82	72.75	21.73	17.31
C3RL	63.22	67.90	27.80	30.45	43.44	62.19	45.54	48.00
+ CalibSFT	63.32	80.88	15.65	15.18	44.60	63.03	23.89	17.38

5.2 OVERALL RESULTS

CalibSFT consistently benefits diverse confidence-aware RL methods. As shown in Table 1, initializing RL from CalibSFT reduces Brier score and ECE and increases AUROC across all five confidence-aware RL baselines on both mathematical and general-reasoning benchmarks. As a concrete example, CalibSFT reduces DCPO’s ECE from 44.59 to 20.25 while lifting its AUROC from 64.19 to 74.73 on MinervaMath. Moreover, these calibration gains come without sacrificing accuracy. The average Pass@1 improves by 1.19 points in-distribution and 1.49 points out-of-distribution across the five methods. We provide detailed per-dataset results in Table 6-7 (Appendix E.1). Overall, these results establish CalibSFT as a plug-and-play initialization that consistently strengthens confidence-aware RL for better calibration.

CalibSFT and RL contribute complementary gains. To disentangle their contributions, we compare CalibSFT and RLVR individually. CalibSFT alone substantially mitigates overconfidence, with ECE dropping from 36.26 to 14.82 in-distribution and from 46.94 to 24.16 out-of-distribution. In contrast, Base and RLVR remain severely miscalibrated, indicating that answer correctness alone is insufficient to reshape confidence. However, calibration does not imply better confidence ranking: CalibSFT’s OOD AUROC even declines slightly from 65.71 to 61.38, reflecting the limited generalization of supervised training. Confidence-aware RL fills exactly this gap by directly optimizing discrimination, and combining it with CalibSFT yields the strongest overall performance.

5.3 CALIBRATION ANALYSIS

CalibSFT aligns confidence with accuracy across tasks. A well-calibrated model should express higher confidence on tasks it can reliably solve and lower confidence on harder ones. To test this, we compute Pearson and Spearman correlations between per-dataset mean confidence and per-dataset accuracy for each RL method on the ID and OOD splits separately. As shown in Table 2, CalibSFT improves Spearman correlation for all RL baselines on both mathematical and general reasoning benchmarks. For example, DCPO’s Spearman correlation rises from 0.619 to 0.976 on mathematical tasks and from 0.548 to 0.643 on general reasoning. Pearson correlation also improves consistently across all baselines and splits (e.g., from 0.505 to 0.729 for RLCR on OOD). These results indicate that CalibSFT helps the model assign higher confidence to tasks it is more likely to answer correctly.

CalibSFT mitigates downstream confidence concentration after RL. Prior work observes that confidence estimates often concentrate on a few dominant values even after confidence-aware RL training (Wang et al., 2026; Yang et al., 2026). To examine whether CalibSFT initialization alleviates

Table 2: Pearson r and Spearman ρ correlation between accuracy and mean confidence across ID and OOD datasets. Both correlations increase for all RL methods when incorporating CalibSFT, indicating that confidence better tracks task-level accuracy.

Dataset	Method	RLCR		ReDoubt		CoCA		DCPO		C3RL	
		r	ρ	r	ρ	r	ρ	r	ρ	r	ρ
ID	Baseline	0.768	0.619	0.829	0.762	0.750	0.643	0.669	0.619	0.635	0.595
	+CalibSFT	0.963	0.952	0.976	0.976	0.887	0.976	0.996	0.976	0.896	0.833
OOD	Baseline	0.505	0.548	0.667	0.762	0.447	0.571	0.467	0.548	0.681	0.714
	+CalibSFT	0.729	0.619	0.738	0.881	0.605	0.619	0.611	0.643	0.848	0.929

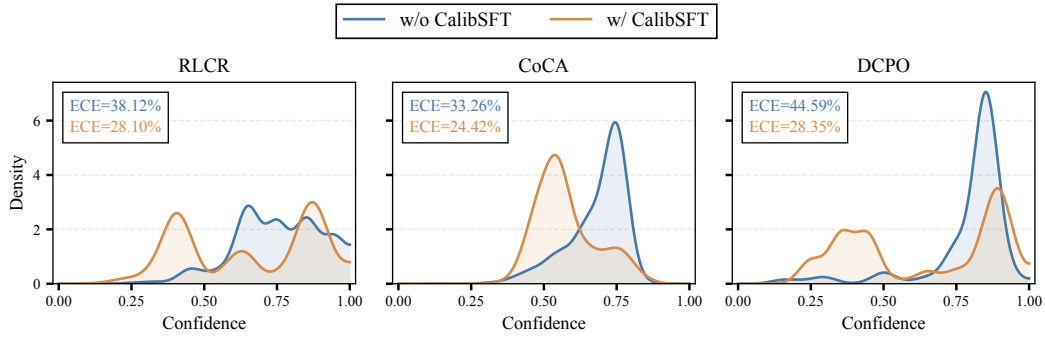


Figure 4: Confidence distributions on MinervaMath for RLCR, CoCA, and DCPO. CalibSFT reduces concentration at dominant confidence values and lowers ECE for all three methods.

this concentration, we compare RLCR, CoCA, and DCPO on MinervaMath, with and without CalibSFT. As shown in Figure 4, all three methods trained from Base exhibit heavy peaks at a few high-confidence values, while initializing from CalibSFT flattens these peaks and aligns the mean confidence with the actual accuracy. As a result, CalibSFT alleviates both concentration and overconfidence and lowers ECE for all three methods. For example, DCPO’s dominant peak near high confidence is redistributed toward lower values while accuracy is preserved, lowering ECE from 44.59% to 28.35%. These results suggest that CalibSFT initialization not only calibrates the SFT stage but also propagates lower concentration and better calibration into the downstream RL policy.

The calibration gains of CalibSFT persist across reasoning budgets.

A useful calibrator should keep confidence reliable regardless of how much test-time reasoning the model is allocated. To test this, we compare RLCR with CalibSFT on AIME 2026 across a range of token budgets. As shown in Figure 5, CalibSFT achieves substantially lower ECE at every budget while accuracy remains comparable to the RLCR baseline. For example, at a 10,000-token budget, ECE drops from 22.71 to 10.97 without sacrificing accuracy. These results indicate that CalibSFT’s calibration gains persist across reasoning budgets.

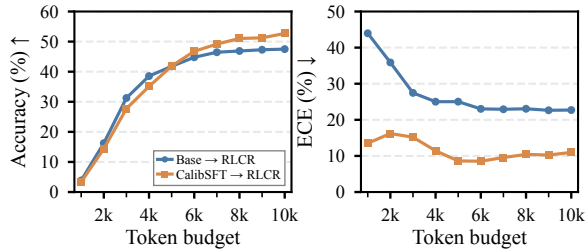


Figure 5: Accuracy (↑) and ECE (↓) on AIME’26 across token budgets for RLCR. CalibSFT initialization yields lower ECE across all budgets while maintaining accuracy.

Table 3: AUROC ($\uparrow$) and ECE ($\downarrow$) of Qwen3-8B across confidence formats. “Base” denotes the model without training and “+Ours” the model after CalibSFT. Values are averaged over the eight ID or eight OOD datasets.

Confidence Format	ID				OOD			
	ECE $\downarrow$		AUROC $\uparrow$		ECE $\downarrow$		AUROC $\uparrow$	
	Base	+Ours	Base	+Ours	Base	+Ours	Base	+Ours
Probability $\{0.00, 0.01, \dots, 1.00\}$	16.64	10.83	82.05	86.88	32.81	18.20	68.72	71.31
Probability $\{0.00, 0.05, \dots, 1.00\}$	16.64	14.39	82.05	86.77	32.81	19.30	68.72	71.23
Integer (0–9)	16.91	15.84	78.86	84.85	26.31	16.70	67.00	69.36
Linguistic (low / medium / high)	17.71	15.90	73.56	79.58	24.06	20.31	64.20	67.73

Table 4: AUROC ($\uparrow$) and ECE ($\downarrow$) of RLCR on Gemma4-E2B across benchmarks of varying difficulty. CalibSFT noticeably improves calibration across all difficulty levels.

Metric	Method	In-distribution			Out-of-distribution			Average
		Math500	Olympiad	Minerva	DROP	HotpotQA	NQOpen	
ECE $\downarrow$	RLCR	8.09	22.63	40.72	19.92	26.41	38.25	26.00
	+CalibSFT	5.44	12.73	20.74	13.30	13.04	22.77	14.67
AUROC $\uparrow$	RLCR	74.96	83.64	67.71	60.60	60.48	76.85	70.71
	+CalibSFT	80.23	88.20	71.44	65.45	62.66	77.13	74.69

5.4 GENERALITY OF CALIBSFT

CalibSFT is effective across diverse confidence formats. We further show that CalibSFT is effective across different ways of expressing confidence. Specifically we train CalibSFT with four confidence target formats: two-decimal probabilities on a fine grid $\{0.00, 0.01, \dots, 1.00\}$ or a coarse grid $\{0.00, 0.05, \dots, 1.00\}$, integer scores from 0 to 9, and linguistic labels (*low*, *medium*, *high*) (See Appendix D.3 for details). As shown in Table 3, CalibSFT consistently improves both ECE and AUROC across all four formats, indicating that its benefit does not rely on a particular way of expressing confidence. The reduction in ECE is largest with the fine probability grid, whose granularity lets confidence approximate the probability of correctness most closely.

CalibSFT is agnostic to the base model. To examine the transferability of CalibSFT across model architectures, we train Gemma4-E2B (Abd et al., 2026) using RLCR with CalibSFT, and then evaluate it on three ID and three OOD benchmarks spanning a range of difficulty. As shown in Table 4, CalibSFT improves both ECE and AUROC on all benchmarks, halving the average ECE from 26.00 to 14.67 and raising the average AUROC from 70.71 to 74.69. These results indicate that the calibration benefit of CalibSFT holds across model families on benchmarks of varying difficulty.

6 CONCLUSION

[TODO:] Summarize the concentrated and misaligned confidence prior, the rollout-derived and target-balanced CalibSFT initialization, and its effect on downstream confidence-aware RL. Close with the main ID/OOD evidence and the remaining limitations of rollout-derived confidence targets.

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AUTHOR CONTRIBUTIONS

If you’d like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

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A RELATED WORK

Confidence estimation for LLMs. Confidence estimates for a model answer can be derived from many signals, of which four families cover most existing methods and differ in what they require at inference time (Geng et al., 2024; Liu et al., 2025). The most direct signal is the probability the model assigns to its own answer tokens, used as the confidence either as it is or after a layer fitted on labeled data corrects it (Kadavath et al., 2022; Liu et al., 2024b). Probing methods instead train a classifier on the hidden states or on the layer-wise trajectory that produces the answer (Azaria & Mitchell, 2023; Eusebi et al., 2026; Srey et al., 2026), which requires access to activations rather than logits. When a model exposes neither, sampling-based methods recover confidence from behavior alone and measure how far several sampled answers agree (Kuhn et al., 2023; Del et al., 2026; Taubenfeld et al., 2025), at the cost of repeated generations per question. Verbalized confidence removes that cost as well: the model states a numerical or linguistic score alongside its answer within a single generation (Lin et al., 2022; Tian et al., 2023; Xiong et al., 2024; Yang et al., 2024). We study the verbalized setting for reasoning models, where the score follows a reasoning trace and the final answer. Such a score is informative only once it matches accuracy: the values an untrained model reports concentrate on a few points of the scale (Dai & Wang, 2026; Cacioli, 2026) and rank its errors worse than its own token probabilities (Ni et al., 2024; Tao et al., 2025).

Confidence calibration for LLMs. Calibration requires the reported confidence to match the empirical probability that the answer is correct, and a method either rescales a score after the model has produced it or changes the model that produces it. Rescaling is the standard treatment for probability-based signals: a temperature or a bias layer is fitted on a labeled held-out set (Guo et al., 2017; Liu et al., 2024b), and label-free variants align the score with a reference model or project the two predictive distributions onto a calibrated one (Luo et al., 2026; Kong et al., 2026). Probes need no separate step, since they are classifiers already trained on correctness labels (Azaria & Mitchell, 2023; Srey et al., 2026), and a sampling-based estimate is calibrated by choosing how agreement among samples maps to a probability (Kuhn et al., 2023; Del et al., 2026). A verbalized score can be rescaled the same way, but only through extra queries per question or an extra pass that rewrites the reported value (Wang & Stengel-Eskin, 2026; Yeh et al., 2026), and neither changes how the model generates the score in the first place.

Post-training instead changes the model that emits the score, and it targets the verbalized score because that score needs no logit access and no extra samples at deployment. Supervised fine-tuning supervises the score with correctness signals collected from the model’s own responses (Lin et al., 2022; Zhang et al., 2024; Jang et al., 2025; Liu et al., 2024a; Xu et al., 2024; Hager et al., 2025; Seo et al., 2025; Zhang et al., 2026a; Li et al., 2025). Reinforcement learning adds a calibration term to the reward of RLVR, which by itself optimizes answer correctness and leaves the confidence unsupervised (Shao et al., 2024; Yu et al., 2025); the term is a proper scoring rule on the reported confidence, and the methods differ in where the score is placed, what it is trained toward, and how credit reaches its tokens (Leng et al., 2025; Damani et al., 2026; Bani-Harouni et al., 2026; Tan et al., 2026; Li et al., 2026a; Finlay et al., 2026; Shi et al., 2026; Yu et al., 2026; Li et al., 2026b; Ma et al., 2026; Wang et al., 2026; Yang et al., 2026; Zhang et al., 2026b). These objectives learn from the confidence values the base model already samples, which Section 3 shows are concentrated on a few high values and stay concentrated while the calibration reward is optimized. In this paper, we analyze this prior and propose CalibSFT, an SFT stage that initializes verbalized confidence before confidence-aware RL.

B THE ANALYSIS OF THE MOTIVATION

In Section 3, we observe that confidence stays concentrated on a few values while RLVR optimizes its calibration reward. To verify that this behavior is not specific to one objective, we further train Qwen3-8B with CoCA (Li et al., 2026a) and DCPO (Ma et al., 2026) under the same setup and track the same quantities.

As shown in Figure 6, we observe the same phenomenon as in Section 3. Between the first and last ten steps, Brier loss falls for RLVR (0.266 to 0.224) and CoCA (0.255 to 0.221) and stays nearly flat for DCPO (0.246 to 0.241), so each objective either improves calibration or holds it steady. Confidence support does not follow. The mean number of distinct confidence values within a group of eight

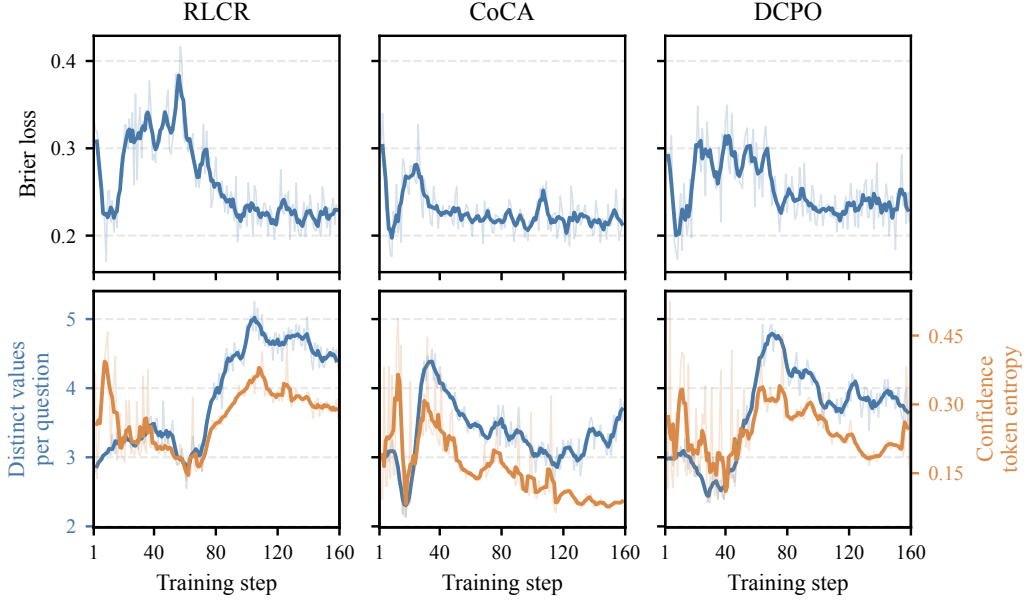


Figure 6: Training dynamics of confidence calibration and support for Base→RLCR, Base→CoCA, and Base→DCPO. The top row shows Brier loss; the bottom row shows distinct confidence values per question (blue, left axis) and confidence token entropy (orange, right axis). Faint curves show per-step measurements and bold curves centered five-step means. Statistics use the groups retained by the corresponding training run, so the sampled questions and responses change across steps.

responses never exceeds 5.3 at any step of any run, and confidence token entropy ends below where it started in all three runs. Therefore, none of the three objectives moves the model away from sampling a few confidence values per question, which extends the observation reported for RLCR in Section 3 to all three objectives.

C THEORETICAL ANALYSIS

C.1 PROOF OF PROPOSITION 1

Proof. We fix x and the expected rewards u_i , and differentiate with respect to the confidence logits. The softmax derivative is $\partial\pi_j/\partial\eta_i = \pi_j(\mathbf{1}\{j=i\} - \pi_i)$. It follows that

$$\frac{\partial J_x}{\partial\eta_i} = \sum_j u_j \pi_j (\mathbf{1}\{j=i\} - \pi_i) \quad (11)$$

$$= \pi_i (u_i - J_x). \quad (12)$$

Since J_x is a convex combination of the u_i values, $|u_i - J_x| \leq \Delta_x$, which gives the coordinate-wise bound. If $\pi_k = 1 - \varepsilon$, then

$$|u_k - J_x| = \left| \sum_{i \neq k} \pi_i (u_k - u_i) \right| \leq \varepsilon \Delta_x. \quad (13)$$

Consequently,

$$\|\nabla_{\eta} J_x\|_1 = \pi_k |u_k - J_x| + \sum_{i \neq k} \pi_i |u_i - J_x| \quad (14)$$

$$\leq (1 - \varepsilon) \varepsilon \Delta_x + \varepsilon \Delta_x \quad (15)$$

$$\leq 2\varepsilon \Delta_x. \quad (16)$$

□

The gradient is the expectation of the likelihood-ratio estimator $U \nabla_{\eta} \log \pi_I$, where $I \sim \pi$ and $U = -(c_I - Z)^2$, with Z drawn from its fixed conditional distribution given x , independently of I . This matches RLGR's reward $Z - (c_I - Z)^2$ (Preliminaries) up to the term Z , which does not depend on c_I . Assuming $C \perp Z \mid x$ as in Proposition 2, this term contributes an η -independent additive constant to J_x that does not affect this gradient. A baseline depending only on x leaves this expectation unchanged.

C.2 PROOF OF PROPOSITION 2

Proof. Conditional independence gives $\mathbb{E}[Z - p_x \mid x, C] = 0$. Expanding the square therefore yields

$$\mathbb{E}[(C - Z)^2 \mid x] = \mathbb{E}[(Z - p_x)^2 \mid x] + \mathbb{E}[(C - p_x)^2 \mid x] \quad (17)$$

$$= p_x(1 - p_x) + \mathbb{E}[(C - p_x)^2 \mid x]. \quad (18)$$

For any $\delta > 0$, the second term satisfies

$$\mathbb{E}[(C - Z)^2 \mid x] - p_x(1 - p_x) = \mathbb{E}[(C - p_x)^2 \mid x] \quad (19)$$

$$\geq \delta^2 \Pr(|C - p_x| \geq \delta \mid x) \quad (20)$$

$$= \delta^2(1 - m_{\delta}(x)). \quad (21)$$

Thus, assigning little probability mass near p_x incurs excess Brier loss even when confidence values in that neighborhood remain possible. $\square$

Restricted confidence support. As a special case, suppose $\Pr(C \in S_x \mid x) = 1$ for a nonempty set $S_x \subseteq [0, 1]$, and define $d_x = \text{dist}(p_x, S_x) = \inf_{c \in S_x} |c - p_x|$. If $d_x > 0$, then $m_{d_x}(x) = 0$, and Proposition 2 with $\delta = d_x$ gives

$$\mathbb{E}[(C - Z)^2 \mid x] \geq p_x(1 - p_x) + \text{dist}(p_x, S_x)^2. \quad (22)$$

For $d_x = 0$, the same bound follows directly from the decomposition above. If S_x is closed, a predictor that always outputs a nearest point in S_x attains the bound. Confidence values are effectively drawn from a small discrete set throughout this paper (Figure 1), so $\text{dist}(p_x, S_x)$ measures how far the nearest available value falls from p_x , translating the observed concentration directly into excess Brier risk.

Best-of- G sampling under a concentrated policy. Given the policy and a question x , draw G independent confidence samples $C_1, \dots, C_G$ from its conditional distribution. Each sample lies outside $\{c : |c - p_x| < \delta\}$ with probability $1 - m_{\delta}(x)$, and by independence all G samples do so with probability $(1 - m_{\delta}(x))^G$. On that event, even the sample closest to p_x has squared error at least δ^2 , so

$$\mathbb{E} \left[\min_{i \leq G} (C_i - p_x)^2 \mid x \right] \geq \delta^2(1 - m_{\delta}(x))^G. \quad (23)$$

With the group size $G = 8$ used in our RL runs, a policy that rarely samples near p_x is unlikely to produce a well-calibrated confidence anywhere in the group.

C.3 PROOF OF PROPOSITION 3

We analyze the unrounded targets under the original sampling distribution, before response filtering and target balancing. The random variables Q_x and C_i^* correspond to the observed q_x and c_i^* in Equation 8.

Proof of Proposition 3. Fix x and write $v_x = p_x(1 - p_x)$. Conditional independence gives

$$\mathbb{E}[Q_x \mid x] = \mathbb{E}[Z_i \mid x] = p_x, \quad \text{Var}(Q_x \mid x) = \frac{v_x}{n}. \quad (24)$$

Since Z_i is included in Q_x , their conditional covariance is

$$\text{Cov}(Q_x, Z_i \mid x) = \frac{v_x}{n}. \quad (25)$$

Linearity of expectation gives $\mathbb{E}[C_i^* | x] = p_x$. Expanding the variance yields

$$\text{Var}(C_i^* | x) = \lambda^2 \frac{v_x}{n} + (1 - \lambda)^2 v_x + 2\lambda(1 - \lambda) \frac{v_x}{n} \quad (26)$$

$$= v_x \left[\frac{1}{n} + (1 - \lambda)^2 \left(1 - \frac{1}{n} \right) \right]. \quad (27)$$

Now fix a at which ℓ is finite. Since $\ell(a, t)$ is affine in t , unbiasedness gives

$$\mathbb{E}[\ell(a, C_i^*) | x] = p_x \ell(a, 1) + (1 - p_x) \ell(a, 0) = \mathbb{E}[\ell(a, Z_i) | x]. \quad (28)$$

Because ℓ is strictly proper, this expected loss is uniquely minimized at $a = p_x$. Thus the mixture, the group average, and an individual correctness label induce the same expected scoring loss for a prediction based on x . For the Brier loss $\ell(a, z) = (a - z)^2$, the extension is $\ell(a, t) = (a - t)^2 + t(1 - t)$, so the squared loss against a soft target has the same minimizer. The same argument applies to the binary logarithmic loss (Gneiting & Raftery, 2007). $\square$

Effect of the response count and mixing weight. At $\lambda = 0$, the target is the binary label with variance v_x ; at $\lambda = 1$, it is the group average with variance v_x/n . For a fixed $\lambda < 1$, the variance approaches $(1 - \lambda)^2 v_x$ as n increases, reflecting the retained contribution of the individual outcome. Within any observed group, $C_i^* - C_j^* = (1 - \lambda)(Z_i - Z_j)$, so the targets of a correct and an incorrect response differ by $1 - \lambda$.

Conditioning on the generated response. Proposition 3 concerns the success probability p_x before observing a particular response. To make the effect of conditioning explicit, let h_i include x and information from response i , and define $p_{h_i} = \Pr(Z_i = 1 | h_i)$. If the other responses are independent of h_i conditional on x , then

$$\mathbb{E}[C_i^* | h_i] = \left(1 - \lambda + \frac{\lambda}{n} \right) p_{h_i} + \lambda \left(1 - \frac{1}{n} \right) p_x. \quad (29)$$

The target thus shrinks the response-conditional probability toward the question’s success probability. These results justify the mean and variance of this supervision signal under the original sampling distribution; it does not establish response-conditional calibration after target balancing and token-level SFT. Those effects are evaluated empirically.

D IMPLEMENTATION DETAILS

D.1 TRAINING PIPELINES

All runs use Qwen3-8B as the base model. From DeepScaleR-Preview we remove duplicate questions and questions whose gold answers disagree across duplicates, which leaves 38,915 questions. We randomly hold out 2,000 of them as a test split (DeepScaleR-Eval) and use the remaining 36,915 as the training split (DeepScaleR-Train), which is used to construct the SFT data for CalibSFT and the downstream RL training data.

SFT data construction. We sample $n = 50$ responses per question from DeepScaleR-Train with the base model, using temperature 1.0, top- p 0.95, and a cap of 8,192 tokens, which gives 1,845,750 responses in total. Each question’s success rate q_x is computed from all 50 responses before any filtering. We then discard responses that do not follow the required `<think>/<answer>/<confidence>` format and responses that hit the token cap without terminating, which are typically repetitive or looping generations. This removes 401,634 responses and retains 1,444,116, with q_x kept fixed. For each retained response we form the target $c_i^* = \lambda q_x + (1 - \lambda) z_i$ with $\lambda = 0.5$ and replace the confidence suffix with it, preserving the sampled reasoning and answer. In the main probability format, the target is rendered to two decimal places in the trailing `<confidence>` tag, which yields 100 distinct target values. For balanced sampling, we group examples by the numeric target rounded to six decimal places and sample the same number from every nonempty group, using the smallest group size as the default count. We apply this rule either through a per-epoch sampler or when constructing an offline balanced split; the offline split used in our main experiments contains 349 examples per target value, that is, 34,900 examples drawn from 11,539 distinct questions.

CalibSFT. We train on the balanced split with the correctness-conditional loss mask and $\beta = 0.5$: correct responses supervise both the reasoning/answer and the confidence segments, whereas incorrect responses supervise only the confidence segment. We use AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with no weight decay and gradient-norm clipping at 1.0, a constant learning rate of 2×10^{-6} after 10 warm-up steps, and a global batch size of 256 sequences truncated to 16,384 tokens. We train for 180 optimization steps, roughly 1.3 epochs over the balanced split, on eight A100 GPUs.

Confidence-aware RL. For downstream RL, each prompt group contains 8 sampled responses, with each response capped at 8,192 tokens. We train with batches of 64 prompt groups for 160 optimization steps. For methods using DAPO dynamic sampling, this corresponds to approximately 600 generation batches. Dynamic sampling retains only groups with mixed correctness outcomes and discards groups whose responses are all correct or all incorrect. To accelerate training for these methods, we remove questions for which all sampled responses from the base model are correct before RL training begins. Each method retains its own reward and credit assignment.

- **Direct RL baselines.** RLVR, RLCR (Damani et al., 2026), CoCA (Li et al., 2026a), and DCPO (Ma et al., 2026) are trained directly from the Qwen3-8B base checkpoint. They share the DAPO training recipe but retain their respective correctness, calibration-target, and credit-assignment objectives.
- **ReDoubt.** ReDoubt is initialized from the trained RLVR checkpoint and then optimized with its clipped log-score confidence objective (Bani-Harouni et al., 2026). When incorporated with CalibSFT, we first train CalibSFT on the RLVR checkpoint and then continue with ReDoubt RL.
- **C3RL.** C3RL (Yang et al., 2026) combines correctness, calibration, and reference-accuracy rewards. C3RL uses the digit confidence format, expressing confidence as an integer from 0 to 9 and mapping it to $[0, 1]$ by dividing by 9. The CalibSFT initialization for C3RL is also trained in this format. We use the base model’s success rates over sampled responses to label each training question as all correct, partially correct, or all incorrect. The reference reward encourages correct answers to previously all-incorrect questions and penalizes incorrect answers to previously all-correct questions. To retain these questions and their reference-reward signals, we disable DAPO dynamic sampling and all-correct filtering, using the full RL training split for C3RL both with and without CalibSFT initialization.

D.2 EVALUATION PROTOCOL

For the main results, we use temperature 0.6 and sample four responses per question on non-AIME mathematical and OOD benchmarks and 32 responses per question on AIME. Pass@1 is the mean correctness over these sampled responses. The supervision and target ablations retain their recorded evaluation settings. Confidence is parsed from the final probability field and normalized to $[0, 1]$. We use 10 equal-width bins for calculating ECE. Unless otherwise noted, RL results use the final checkpoint after 160 optimization steps, while SFT-only results use the checkpoint at the end of their respective SFT runs.

Answer verification. Every benchmark is scored by a deterministic verifier except TriviaQA. On the mathematical benchmarks and HotpotQA, a response is correct if its extracted answer is symbolically equivalent to the gold answer under `math_verify`, or if the two agree after lowercasing, removing punctuation and articles, and collapsing whitespace. NQOpen and MuSiQue apply the same normalized match against any gold alias. PopQA follows its official protocol and accepts a response that contains a gold alias, and WebQuestions requires the set of predicted answer items to match the gold set exactly. DROP uses the official DROP normalization and counts a response correct only on an exact match with one of its references. LiveBenchReasoning uses LiveBench’s own deterministic graders for its spatial, web-of-lies, and zebra-puzzle tasks. TriviaQA admits many surface forms per answer, so we score it with an LLM judge: Qwen3-8B receives the question, the reference answer, and the candidate answer, and returns a correct or incorrect verdict. The same symbolic-equivalence-or-normalized-match rule produces the correctness labels z_i that build the CalibSFT targets and the correctness rewards during RL.

D.3 CONFIDENCE FORMATS

The format comparison in Section 5.4 keeps the CalibSFT training data fixed and changes only how the confidence targets are written. Probability targets use two decimal places, and we compare two target grids with steps of 0.01 and 0.05. Integer targets map a target c^* to $\lfloor 9c^* + 0.5 \rfloor \in \{0, \dots, 9\}$, and linguistic targets assign *low*, *medium*, and *high* to $c^* < 1/3$, $1/3 \leq c^* < 2/3$, and $c^* \geq 2/3$, respectively. Table 5 lists the system prompt used for each format. For evaluation, an integer d is mapped to $d/9$ and the labels *low*, *medium*, and *high* to 0.25, 0.5, and 0.75.

Table 5: System prompts for the three confidence formats.

Format	System prompt
Probability	A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and analyzes its confidence about the solution and then provides the user with the final answer as well as its confidence level. The confidence level indicates how certain the Assistant is about its answer, expressed as a decimal between 0 and 1 with exactly two decimal places, enclosed within <code><confidence></code> <code></confidence></code> tags. The response must strictly follow this format: <code><think></code> reasoning process here <code></think></code> <code><answer></code> final short answer only <code></answer></code> <code><confidence></code> 0.xx <code></confidence></code> . The <code><answer></code> tag must contain only the final answer string needed for exact-match evaluation, not a full sentence, explanation, or reasoning.
Integer	A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and analyzes its confidence about the solution and then provides the user with the final answer as well as its confidence score. The confidence score indicates how certain the Assistant is about its answer, expressed as a single integer from 0 to 9, enclosed within <code><confidence></code> <code></confidence></code> tags. A score of 0 means very low confidence and the answer is likely incorrect. A score of 9 means very high confidence and the answer is very likely correct. The response must strictly follow this format: <code><think></code> reasoning process here <code></think></code> <code><answer></code> final short answer only <code></answer></code> <code><confidence></code> X <code></confidence></code> , where X is one of 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. The <code><answer></code> tag must contain only the final answer string needed for exact-match evaluation, not a full sentence, explanation, or reasoning.
Linguistic	A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and analyzes its confidence about the solution and then provides the user with the final answer as well as its confidence level. The confidence level indicates how certain the Assistant is about its answer, expressed as exactly one of low, medium, or high, enclosed within <code><confidence></code> <code></confidence></code> tags. Low means the answer is likely incorrect, medium means the answer is uncertain, and high means the answer is likely correct. The response must strictly follow this format: <code><think></code> reasoning process here <code></think></code> <code><answer></code> final short answer only <code></answer></code> <code><confidence></code> label <code></confidence></code> . The <code><answer></code> tag must contain only the final answer string needed for exact-match evaluation, not a full sentence, explanation, or reasoning.

E DETAILED EXPERIMENTAL RESULTS

E.1 PER-DATASET MAIN RESULTS

Tables 6 and 7 report the results on each individual benchmark, complementing the split-level averages in Table 1. We present the average over its eight datasets in the last subtable of each split.

The improvements hold across benchmarks. The gains are largest where the uninitialized RL models are most overconfident. On the six benchmarks with baseline ECE above 25 (MinervaMath, MuSiQue, LiveBenchReasoning, NQOpen, PopQA, and WebQuestions), CalibSFT initialization reduces both Brier score and ECE for all methods. The Base, RLVR, and CalibSFT-alone rows show what each stage contributes. RLVR raises Pass@1 but leaves calibration unchanged, and on the out-of-distribution split it even worsens Brier score and ECE relative to Base, matching the failure mode in Section 3. CalibSFT alone substantially reduces Brier score and ECE on both splits and improves

Table 6: Per-dataset results on the in-domain mathematical benchmarks. Base, RLVR, and standalone CalibSFT use 6,144 tokens; downstream RL models use 16,384 tokens. Temperature is 0.6, with four responses per non-AIME question and 32 per AIME question.

(a) DeepScaleR-Eval					(b) MATH-500					(c) MinervaMath				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	60.84	66.31	34.09	34.06	Base	83.90	63.80	14.78	13.87	Base	39.89	63.26	53.79	55.28
RLVR	70.15	68.11	26.94	26.37	RLVR	89.15	67.69	10.13	8.76	RLVR	37.96	60.92	57.18	58.40
CalibSFT	60.88	73.76	23.60	17.57	CalibSFT	83.65	82.11	14.70	11.53	CalibSFT	39.89	57.50	30.33	24.20
RLCR	71.24	75.33	16.65	5.14	RLCR	90.50	76.39	7.54	7.43	RLCR	36.67	66.67	35.91	38.12
+ CalibSFT	72.47	79.34	15.33	4.58	+ CalibSFT	90.15	89.79	6.56	3.69	+ CalibSFT	37.04	76.29	26.81	26.33
ReDoubt	71.53	75.92	17.13	10.28	ReDoubt	90.05	74.85	9.82	15.01	ReDoubt	37.59	66.65	27.92	25.37
+ CalibSFT	72.36	78.06	16.06	6.67	+ CalibSFT	89.45	89.64	9.60	16.74	+ CalibSFT	36.21	83.73	21.06	22.86
CoCA	70.16	73.17	17.73	8.51	CoCA	89.85	72.69	10.64	18.18	CoCA	35.20	65.28	32.43	33.26
+ CalibSFT	71.45	78.71	16.41	7.92	+ CalibSFT	89.85	87.29	9.09	15.49	+ CalibSFT	36.95	78.62	23.47	24.42
DCPO	71.45	71.74	17.30	6.50	DCPO	89.30	70.77	7.90	6.27	DCPO	34.93	64.19	41.78	44.59
+ CalibSFT	70.46	77.71	16.06	5.95	+ CalibSFT	90.20	84.89	7.68	7.70	+ CalibSFT	36.86	74.73	23.33	20.25
C3RL	70.65	62.87	24.64	25.05	C3RL	89.40	60.22	9.42	9.75	C3RL	37.22	56.00	58.42	59.49
+ CalibSFT	71.05	71.91	19.93	14.72	+ CalibSFT	87.70	84.76	10.37	9.36	+ CalibSFT	35.85	77.64	20.14	16.95
(d) OlympiadBench					(e) GSM8K					(f) AIME2024				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	54.41	69.82	38.93	40.13	Base	94.35	67.82	5.20	3.39	Base	48.23	81.93	39.61	42.50
RLVR	56.31	72.84	39.35	39.98	RLVR	94.64	67.97	5.06	3.05	RLVR	52.92	82.05	39.49	41.15
CalibSFT	51.67	79.33	22.48	20.62	CalibSFT	94.31	70.24	17.90	20.78	CalibSFT	45.42	89.96	11.93	6.03
RLCR	57.83	81.78	19.04	12.50	RLCR	94.81	78.88	4.77	5.58	RLCR	59.48	93.59	13.17	19.70
+ CalibSFT	58.46	89.71	16.09	13.25	+ CalibSFT	94.67	82.60	4.64	2.72	+ CalibSFT	64.17	92.42	10.97	9.79
ReDoubt	57.57	80.67	18.18	8.36	ReDoubt	94.96	78.40	7.07	16.27	ReDoubt	55.62	91.94	15.40	22.77
+ CalibSFT	58.35	91.01	14.96	16.88	+ CalibSFT	94.69	80.28	7.50	17.62	+ CalibSFT	61.56	93.81	11.89	14.37
CoCA	56.38	80.19	20.14	14.63	CoCA	94.58	69.00	8.94	21.02	CoCA	53.85	92.22	16.27	21.37
+ CalibSFT	56.19	90.47	18.09	20.16	+ CalibSFT	94.56	71.85	8.26	18.10	+ CalibSFT	58.54	95.41	16.34	23.56
DCPO	57.05	78.43	21.85	17.75	DCPO	94.48	68.34	5.50	9.13	DCPO	57.92	88.84	12.32	8.47
+ CalibSFT	57.46	88.53	15.72	12.14	+ CalibSFT	94.45	73.40	5.80	7.48	+ CalibSFT	58.33	93.89	11.02	11.85
C3RL	57.68	62.66	35.85	37.52	C3RL	94.37	57.25	5.33	5.24	C3RL	59.69	80.93	25.25	30.19
+ CalibSFT	56.64	84.53	17.91	18.60	+ CalibSFT	94.29	64.91	5.26	5.08	+ CalibSFT	64.27	85.24	14.04	12.99
(g) AIME2025					(h) AIME2026					(i) Overall (ID)				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	35.00	82.75	48.25	52.81	Base	42.50	76.08	45.13	48.07	Base	57.39	71.47	34.97	36.26
RLVR	43.44	84.95	45.87	49.00	RLVR	48.23	80.70	43.82	45.92	RLVR	61.60	73.16	33.48	34.08
CalibSFT	34.27	92.48	9.95	8.02	CalibSFT	39.48	88.65	13.22	9.79	CalibSFT	56.20	79.25	18.01	14.82
RLCR	43.33	95.53	13.60	23.06	RLCR	47.81	94.10	14.08	23.19	RLCR	62.71	82.78	15.60	16.84
+ CalibSFT	48.02	92.40	13.87	14.28	+ CalibSFT	52.60	92.48	13.57	12.04	+ CalibSFT	64.70	86.88	13.48	10.84
ReDoubt	44.90	94.22	15.19	26.51	ReDoubt	52.40	90.93	16.78	26.20	ReDoubt	63.08	81.70	15.94	18.85
+ CalibSFT	51.77	96.32	11.28	16.79	+ CalibSFT	55.52	93.82	11.66	17.69	+ CalibSFT	64.99	88.33	13.00	16.20
CoCA	40.31	95.39	17.04	29.31	CoCA	48.54	96.74	14.84	27.72	CoCA	61.11	80.59	17.25	21.75
+ CalibSFT	45.00	94.21	19.99	26.43	+ CalibSFT	48.65	93.96	19.52	25.63	+ CalibSFT	62.65	86.31	16.40	20.21
DCPO	41.15	91.19	13.33	14.30	DCPO	49.58	91.78	11.04	10.41	DCPO	61.98	78.16	16.38	14.68
+ CalibSFT	41.25	95.07	13.81	17.73	+ CalibSFT	50.21	93.07	12.87	14.52	+ CalibSFT	62.40	85.16	13.29	12.20
C3RL	43.96	79.01	36.25	42.69	C3RL	52.81	84.29	27.21	33.72	C3RL	63.22	67.90	27.80	30.45
+ CalibSFT	43.12	86.74	21.86	25.49	+ CalibSFT	53.65	91.31	15.69	18.30	+ CalibSFT	63.32	80.88	15.65	15.18

Table 7: Per-dataset results on the revised eight-dataset out-of-domain suite. Each question has four sampled responses at temperature 0.6. WebQuestions contains only examples with a single gold answer.

(a) HotpotQA					(b) TriviaQA					(c) DROP				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	64.68	57.68	32.82	32.16	Base	55.24	75.80	35.68	36.93	Base	80.36	59.13	18.22	17.03
RLVR	64.58	54.16	33.10	32.17	RLVR	54.97	76.01	40.59	41.40	RLVR	83.53	59.32	15.34	13.31
CalibSFT	65.70	54.18	29.38	20.57	CalibSFT	55.21	65.05	28.44	20.98	CalibSFT	79.92	59.38	31.60	28.93
RLCR	65.80	59.10	25.97	19.77	RLCR	55.24	81.66	23.84	23.30	RLCR	81.44	63.34	14.72	5.43
+ CalibSFT	66.27	59.23	24.25	10.85	+ CalibSFT	55.25	81.53	18.89	10.71	+ CalibSFT	82.32	74.44	14.99	10.75
ReDoubt	64.83	60.64	23.52	12.11	ReDoubt	55.69	82.96	18.67	13.58	ReDoubt	83.63	63.27	14.76	10.80
+ CalibSFT	64.68	63.08	21.70	1.37	+ CalibSFT	56.14	83.92	19.27	15.58	+ CalibSFT	82.53	75.90	15.60	17.36
CoCA	64.90	56.43	22.77	6.02	CoCA	54.77	77.62	21.63	12.97	CoCA	82.47	61.58	15.03	11.56
+ CalibSFT	66.10	61.57	22.92	11.39	+ CalibSFT	54.89	81.85	20.98	13.16	+ CalibSFT	82.50	73.91	17.95	21.99
DCPO	66.15	57.88	25.33	18.77	DCPO	54.55	79.86	25.93	23.91	DCPO	83.45	59.34	13.25	2.30
+ CalibSFT	65.97	60.70	24.43	14.92	+ CalibSFT	55.12	82.56	17.59	8.25	+ CalibSFT	80.42	75.92	16.54	13.64
C3RL	65.38	52.47	33.39	33.12	C3RL	53.76	73.01	35.61	38.53	C3RL	82.65	54.56	16.37	16.07
+ CalibSFT	66.50	58.94	28.45	19.78	+ CalibSFT	54.11	71.36	22.75	12.50	+ CalibSFT	77.44	64.78	23.65	22.03
(d) MuSiQue					(e) LiveBenchReasoning					(f) NQOpen				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	45.84	62.90	41.91	42.33	Base	52.88	61.81	40.53	40.59	Base	21.12	66.31	66.04	70.59
RLVR	47.18	58.12	46.19	46.75	RLVR	29.25	60.32	61.75	64.59	RLVR	24.13	67.07	68.92	71.55
CalibSFT	47.19	62.25	26.51	13.95	CalibSFT	53.00	67.67	25.40	17.74	CalibSFT	21.64	59.59	27.33	29.16
RLCR	47.77	66.21	25.68	15.35	RLCR	31.75	68.00	28.49	29.09	RLCR	23.32	70.48	43.84	52.43
+ CalibSFT	49.97	67.87	23.22	6.13	+ CalibSFT	42.25	68.41	21.94	7.75	+ CalibSFT	23.80	72.24	26.44	30.89
ReDoubt	46.90	66.37	24.07	11.21	ReDoubt	25.75	68.44	23.51	24.86	ReDoubt	24.67	70.28	32.15	39.19
+ CalibSFT	48.52	70.04	22.42	5.52	+ CalibSFT	42.25	80.49	19.95	12.38	+ CalibSFT	24.10	70.60	26.80	31.99
CoCA	46.69	67.03	23.69	10.18	CoCA	32.12	66.75	25.79	23.64	CoCA	22.76	66.59	35.28	43.26
+ CalibSFT	48.76	66.88	23.56	7.72	+ CalibSFT	41.50	68.03	23.24	7.96	+ CalibSFT	23.72	70.77	24.86	28.69
DCPO	47.34	65.48	28.28	20.53	DCPO	31.13	67.14	32.10	33.12	DCPO	23.69	69.80	46.27	53.69
+ CalibSFT	48.96	70.49	22.75	8.55	+ CalibSFT	37.75	71.77	21.12	8.83	+ CalibSFT	23.58	73.13	22.65	25.97
C3RL	45.55	63.39	39.88	40.04	C3RL	33.00	57.18	53.55	55.46	C3RL	22.43	64.77	63.63	68.49
+ CalibSFT	47.47	66.39	24.46	11.87	+ CalibSFT	44.62	53.33	28.11	22.78	+ CalibSFT	22.08	61.18	22.25	17.54
(g) PopQA					(h) WebQuestions					(i) Overall (OOD)				
Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE	Method	P@1	AUROC	Brier	ECE
Base	20.85	74.94	57.89	64.10	Base	21.88	67.10	68.16	71.80	Base	45.36	65.71	45.15	46.94
RLVR	21.37	72.78	68.14	71.96	RLVR	26.02	65.76	68.29	70.41	RLVR	43.88	64.19	50.29	51.52
CalibSFT	21.00	63.88	24.80	26.68	CalibSFT	23.59	59.02	32.52	35.25	CalibSFT	45.91	61.38	28.25	24.16
RLCR	21.29	76.88	37.71	47.90	RLCR	25.07	69.40	48.86	56.11	RLCR	43.96	69.38	31.14	31.17
+ CalibSFT	21.16	76.55	24.76	32.22	+ CalibSFT	24.63	70.20	31.41	36.33	+ CalibSFT	45.71	71.31	23.24	18.20
ReDoubt	21.37	76.83	26.82	36.23	ReDoubt	26.84	70.22	37.08	43.81	ReDoubt	43.71	69.88	25.07	23.98
+ CalibSFT	21.25	77.87	24.54	32.32	+ CalibSFT	24.91	69.57	29.11	34.59	+ CalibSFT	45.55	73.93	22.42	18.89
CoCA	21.06	73.92	30.84	39.88	CoCA	24.65	64.40	38.35	45.46	CoCA	43.68	66.79	26.67	24.12
+ CalibSFT	20.88	76.99	24.12	31.76	+ CalibSFT	26.06	69.71	25.93	28.47	+ CalibSFT	45.55	71.21	22.95	18.89
DCPO	20.99	77.02	41.10	49.96	DCPO	24.59	68.48	50.52	57.04	DCPO	43.99	68.13	32.85	32.41
+ CalibSFT	21.18	76.52	21.29	28.02	+ CalibSFT	25.61	70.87	27.45	30.32	+ CalibSFT	44.82	72.75	21.73	17.31
C3RL	20.81	71.98	54.60	61.92	C3RL	23.94	60.19	67.27	70.35	C3RL	43.44	62.19	45.54	48.00
+ CalibSFT	20.66	66.98	16.56	12.95	+ CalibSFT	23.94	61.29	24.87	19.55	+ CalibSFT	44.60	63.03	23.89	17.38

in-distribution AUROC. Confidence-aware RL then lifts out-of-distribution AUROC from 61.38 to above 71 for four of the five methods. CalibSFT sets the scale of confidence, and RL sharpens its ranking outside the training domain.

E.2 ABLATION STUDIES

Confidence target construction. The construction of confidence targets in CalibSFT is inspired by proper scoring rules, which serve as a foundation for calibration. To verify the effectiveness of this target design, we compare CalibSFT with targets based on group-level correctness (CSFT) (Jang et al., 2025), token probabilities (UaIT) (Liu et al., 2024a), semantic consistency (SaySelf) (Xu et al., 2024), and rubric-based LLM judgments (LoVeC) (Zhang et al., 2026a), using a shared SFT training set and the same optimization objective.

For a fair comparison, we construct the training data from a shared response pool sampled uniformly across empirical success rates so that the comparison covers both easier and harder questions. For each question in DeepScaleR-Train, we use 10 responses generated by Qwen3-8B, with $z_i \in \{0, 1\}$ denoting correctness and $q_x = \frac{1}{10} \sum_{i=1}^{10} z_i$ the empirical success rate. We select 300 questions at each $q_x \in \{0.1, 0.2, \dots, 0.9\}$ and retain one correct and one incorrect response per question, yielding 5,400 responses from 2,700 questions with balanced correctness at each success rate. SaySelf uses only its correct responses following its SFT setup. On this shared subset, we construct the five confidence targets as follows:

- **CSFT.** Both responses receive the group success rate, $c_i^* = q_x$.
- **UaIT.** $c_i^* = \exp(-\text{NLL}_i)$, where NLL_i is the mean negative log-likelihood of the sampled response tokens under the generating model.
- **SaySelf.** We embed the ten responses to each question with Qwen3-Embedding-8B and cluster them at a cosine similarity threshold of 0.98. Each retained correct response is assigned the frequency of its cluster, $c_i^* = |\mathcal{M}_i|/10$, where $\mathcal{M}_i$ is the cluster containing response i .
- **LoVeC.** An LLM judge scores each response on a 0–10 rubric (10: correct with sound reasoning; 8–9: correct with minor gaps; 5–7: partially correct or ambiguous; 2–4: meaningful but incorrect; 0–1: invalid or clearly wrong), normalized as $c_i^* = s_i/10$.
- **CalibSFT.** Following our main design, the target combines group success rate with response correctness, $c_i^* = 0.5q_x + 0.5z_i$.

All variants initialize from Qwen3-8B and use the same supervision objective: reasoning and answer tokens are supervised on correct responses, while confidence tokens are supervised on all responses. We train for 80 steps (40 for SaySelf) with a batch size of 256 and a learning rate of 2×10^{-6} . Table 8 reports the proportion of training targets in each of ten equal-width confidence intervals across the five methods.

Table 9 reports the AUROC and Brier score of the constructed targets before SFT, together with the resulting models’ average performance on in-distribution mathematical benchmarks after SFT. CalibSFT outperforms all four alternatives on both fronts—achieving a lower Brier score and higher AUROC at construction, which transfers to better downstream performance. These results confirm that grounding target construction in proper scoring rules jointly improves calibration and discrimination.

Target mixing weight. We construct confidence targets as $c_i^* = \lambda q_x + (1 - \lambda)z_i$, where q_x is the group success rate and z_i is the correctness of an individual response. To evaluate the effect of the mixing weight, we compare $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ using the same subset as above and keeping

Table 8: Training-target distribution (%) over ten equal-width bins in the confidence target ablation.

Interval	CSFT	UaIT	SaySelf	LoVeC	CalibSFT
[0.0, 0.1]	11.11	0.00	22.07	27.33	11.11
(0.1, 0.2]	11.11	0.00	14.41	3.78	11.11
(0.2, 0.3]	11.11	0.00	11.93	8.17	11.11
(0.3, 0.4]	11.11	0.00	10.22	0.20	11.11
(0.4, 0.5]	11.11	0.00	9.04	1.37	5.56
(0.5, 0.6]	11.11	0.00	8.26	0.63	11.11
(0.6, 0.7]	11.11	6.48	8.89	0.22	11.11
(0.7, 0.8]	11.11	56.06	6.63	1.54	11.11
(0.8, 0.9]	11.11	35.15	5.56	15.80	11.11
(0.9, 1.0]	0.00	2.31	3.00	40.96	5.56

Table 9: AUROC and Brier score on five confidence target construction methods on ID mathematical benchmarks. CalibSFT achieves the better AUROC and Brier score in both phases.

Target	Pre-SFT targets		Post-SFT predictions	
	AUROC $\uparrow$	Brier $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
CSFT	50.00	31.67	73.46	19.15
UaIT	52.05	33.17	74.72	24.24
SaySelf	–	41.33	57.69	39.38
LoVeC	75.40	24.71	71.11	32.32
CalibSFT	100.00	7.92	79.25	18.01

Table 10: Average calibration performance on eight ID and eight OOD benchmarks across mixing weights. Equal weighting ($\lambda = 0.5$) gives the strongest ID performance.

λ	ID			OOD	
	AUROC $\uparrow$	Brier $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$	
0	68.10	32.49	59.20	38.92	
0.25	78.02	18.83	66.88	29.94	
0.5	79.25	18.01	67.81	30.82	
0.75	75.93	20.00	65.82	28.37	
1	73.29	19.90	57.22	35.72	

Table 11: Ablation of the CalibSFT supervision design on eight ID benchmarks. The variants differ in the supervised responses for each segment and in loss balancing. CalibSFT achieves the best calibration while keeping Pass@1 close to the best.

Variant	Reasoning/answer		Confidence		Segment balanced	ID performance			
	Correct	Incorrect	Correct	Incorrect		Pass@1 $\uparrow$	AUROC $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
Confidence-only	–	–	✓	✓	–	43.56	74.62	20.97	17.43
Full-CE	✓	✓	✓	✓	–	65.49	61.66	29.29	27.65
Full-Balanced	✓	✓	✓	✓	✓	65.46	76.67	19.04	15.87
Correct-only	✓	–	✓	–	✓	66.19	77.58	18.95	16.28
CalibSFT	✓	–	✓	✓	✓	65.79	79.25	18.01	14.82

the SFT settings fixed. At $\lambda = 0$, targets are restricted to 0 and 1; this setting yields the highest Brier score in Table 10. At $\lambda = 1$, correct and incorrect responses to the same question receive identical targets, providing no within-question discrimination signal; post-SFT AUROC is also lower than at $\lambda = 0.5$. Equal weighting achieves the highest AUROC and lowest Brier score on ID benchmarks among the tested settings, so we use $\lambda = 0.5$ as the default.

Correctness-conditional supervision. To verify the effectiveness of CalibSFT design, we ablate three choices in the SFT loss: restricting reasoning and answer supervision to correct responses, supervising confidence on all responses, and balancing the segment losses. Table 11 reports the results. Confidence-only supervision leaves Pass@1 far behind the other variants, showing that reasoning and answer supervision is necessary for accuracy. Without segment balancing, calibration is markedly worse: Full-CE trails Full-Balanced on both AUROC and ECE. Restricting reasoning and answer supervision to correct responses slightly raises accuracy, consistent with incorrect reasoning pulling the model toward incorrect solutions, but restricting confidence supervision to correct responses as well narrows its coverage and weakens calibration. Supervising confidence on all responses while keeping reasoning and answer supervision correct-only recovers this coverage: CalibSFT achieves the best AUROC, Brier score, and ECE among all variants, with Pass@1 close to the best. This combination best balances accuracy, calibration, and confidence diversity, giving a well-calibrated starting point for downstream confidence-aware RL.

E.3 TRAINING STABILITY

[TODO:] Show training curves and report the mean and standard deviation of evaluation metrics across training seeds.

E.4 RESULTS UNDER ADDITIONAL METRICS

[TODO:] Report the same comparisons under three additional families of metrics. (i) Accuracy: Pass@ k in addition to Pass@1, with $k = 4$ on the non-AIME benchmarks and $k = 32$ on AIME. (ii)

Calibration: MacroCE, adaptive ECE, AURC/EAURC, and the Brier decomposition into calibration and refinement terms, together with their definitions, aggregation protocols, and ECE sensitivity to 10, 15, and 20 bins. (iii) Confidence diversity: the number of distinct confidence values, Top-1 and Top-3 confidence mass, confidence entropy, and effective number of values, reported per dataset and as ID/OOD averages.

E.5 COMPARISON WITH ALTERNATIVE CONFIDENCE SIGNALS

[TODO:] Compare post-answer verbalized confidence with token- or sequence-level probability, self-consistency, semantic consistency, and a hidden-state probe. Use the same generated answers whenever the signal permits, and report AUROC, Brier score, ECE, inference cost, and model-access requirements. Discuss which gains arise from CalibSFT and which arise from the choice of confidence signal.

E.6 CONFIDENCE VISUALIZATION

[TODO:] Show reliability diagrams and confidence distributions for models with and without CalibSFT initialization.

E.7 TRAINING EFFICIENCY

[TODO:] Compare CalibSFT and confidence-aware RL using wall-clock time, GPU-hours, optimization steps, and generated rollout/token counts.

E.8 QUALITATIVE CASE STUDIES

[TODO:] Include representative high-confidence errors, low-confidence correct answers, confidence corrections preserved after RL, and failure cases.

F DOWNSTREAM APPLICATIONS

F.1 SELECTIVE ANSWERING AND RISK CONTROL

[TODO:] Rank responses by confidence and report risk–coverage curves, AURC, and accuracy at matched coverage. Compare Base, confidence-aware RL, and CalibSFT-initialized RL to test whether the calibration improvements translate into more reliable abstention decisions.

F.2 CONFIDENCE-GUIDED EFFICIENT REASONING

[TODO:] Use confidence to allocate additional generation budget or route uncertain questions to a stronger inference procedure. Compare against fixed-budget and random-routing baselines at matched token cost, reporting accuracy, calibration, and total generated tokens.